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FLUORINE GENERATION BY GAMMA RADIOLYSIS OF A FLUORIDE SALT MIXTURE

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Fluoride salt mixtures of composition, LiF-BeF_2-ZrF_4-UF_4 (64.5, 30.3, 5.0, 0.13 mol%) were irradiated using 10^7–10^8 R/h gamma radiation to achieve high doses. Under these conditions, F_2 is released from the salt and accumulates as an F_2 overpressure until a 2 mol% damage limit in the crystal is reached. At this point, the rate of F_2 generation is equivalent to the rate of recombination with active metal sites accompanying the radiolytic damage in the salt. The activation energy for the rate of recombination has been found to be 39 kJ/mol; and for the conditions of irradiation used here, no F_2 should be released above 150°C.

1 INTRODUCTION

Much interest has been directed in the past to the radiolysis of halide salts, but most of this attention has been focused on the nature of the defect sites produced in the crystal lattice.[1-5] The radiolysis has been extended to the point where micro bubbles of the halogen gas were observed within the lattice under some conditions of radiation exposure,[2,3] and the concern again centered on the nature of the crystalline material as a result of the radiolysis. These generation reactions can be simply expressed as:

$$LiF + h\nu \rightarrow Li^{\bullet} + F^{\bullet}, \tag{1}$$

$$F^{\bullet} + F^{\bullet} \rightarrow F_2, \tag{2}$$

where $h\nu$ is the gamma photon in this case, $Li^{\bullet}$ is a metal site accompanying the generation of $F^{\bullet}$ atoms. Although recombination of $Li^{\bullet}$ and $F^{\bullet}$ to form LiF is possible, some $F^{\bullet}$ atoms can diffuse and combine via Eq. (2) to form the diatomic halogen gas [other metal fluorides present in the salt mixture are expected to behave similarly to LiF as in Eq. (1)]. As the halogen gas either accumulates within the lattice or escapes to the confinement of a containment vessel, active metal sites accumulate to the point that the back reaction† of $F^{\bullet}$ or F_2 with the metal site is equivalent to that of the generation steps of Eqs. (1) and (2). At this point, no further radiation damage (as noted by chemical products) is apparent.

In spite of all this interest in the behavior of the solid state materials during radiolysis, there is little information regarding the extent of the radiolytic damage under extreme conditions: namely, the amount of gaseous products produced and the extent to which any limits of crystal damage are reached for given amounts of radiation.

†These back reaction mechanisms shall be referred to, collectively, as the fluorine recombination reaction with the salt.

More recently, some concern has been expressed[6] regarding the long-term storage of fluoride salts contaminated with radioactivity, since these materials might, through radiolysis, generate enough gas pressure to compromise the containment system. Of particular interest, is the frozen fuel salt from the Molten Salt Reactor Experiment (MSRE). The composition[7] of this salt was originally LiF-BeF_2-ZrF_4-UF_4 (64.5, 30.3, 5.0, 0.13 mol%); but after being used as the MSRE fuel, it now has considerable amounts of other actinides and fission products which can cause radiolysis of the salt and accompanying release of F_2 gas.

Previous investigators[8] sought to determine the amount of F_2 released upon radiolysis in the MSRE fuel salt. It was shown that F_2 is generated below 150°C; but, at that temperature, the rate of F_2 recombination with the active metal sites left in the crystal lattice became equivalent to its rate of generation so that no release of F_2 from the salt mixture was observed. (Net radiolysis of the fuel salt during reactor operations never occurred because the temperature of operation was $\geqslant$550°C and the recombination rate vastly exceeded the F_2 generation rate.) However, these investigations never explored the question of a limit to the amount of fluorine that would be released under extended conditions since concerns for the long-term storage of the fuel salts had not yet developed.

The current study attempted to address the questions related to long-term storage and sought to determine, at least in a qualitative manner, if there was a limit to the amount of radiation damage that would be sustained for a given dose. Admittedly, the practical concern in a long-term storage facility is the effect of a large dose extended over a very long period of time, i.e., at a relatively low dose rate. Nevertheless, to achieve any information in a practical period of time, it was necessary to compress the time scale and increase the dose rate. For this reason, we sought a very high dose source in order to answer the question of limiting pressures of F_2 generated. Before this work could be completed, the source[9] was shut down for an indefinite period of time leaving the data at a state where more experiments were clearly desirable. Because these data are unique, we believe their presentation at this stage is warranted.

2 EXPERIMENTAL PROCEDURE

2.1 *Materials*

The lithium fluoride used in this series of experiments was obtained from Harshaw Chemical Company as single crystal fragments. The zirconium fluoride and uranium fluoride were from Oak Ridge National Laboratory (ORNL) stock material remaining from the MSRE program. The beryllium fluoride, hydrogen fluoride, and fluorine were obtained from Alfa Products, Linde Specialty Gases, and Air Products, respectively. The graphite used in this work was POCO EDM-3.

All materials of construction were resistant to attack by both hydrogen fluoride and fluorine. Nickel vessels and tubing and monel valves, fittings, and gauges were used where possible. A gas manifold was constructed of resistant materials for use in handling corrosive and toxic gases during sample preparation.

A controlled atmosphere glove box was used for storage and handling of salt mixtures. A helium atmosphere was circulated through molecular sieves (13X Linde) to maintain a moisture content of <10 ppm in the glove box.

2.2 *Methods*

2.2.1 *Fluoride salt preparation* Two fluoride salt mixtures (200 g each) were prepared for irradiation. Their composition was identical to that of the MSRE fuel salt mixture[7] either with or without the uranium fluoride component. The first mixture contained lithium fluoride, beryllium fluoride, and zirconium fluoride (64.6, 30.4, 5.0 mol%); the second contained lithium fluoride, beryllium fluoride, zirconium fluoride, and uranium fluoride (64.5, 30.3, 5.0, 0.13 mol%). Each mixture was weighed into a graphite tube (4.5 cm diam × 15 cm) and loaded into a flanged nickel vessel in a vertical tube furnace.

A Ti-sponge-purified (at 600°C) helium purge was used to sweep the vessel and associated lines while the furnace temperature was increased to 550°C, thus melting the mixture. This is approximately 100°C above the liquidus point of the salt mixture.[7] A graphite dip leg was inserted into the melt and hydrogen fluoride was added to the helium carrier and bubbled (or "sparged") through the molten salt mixture.

This hydrofluorination process removed traces of moisture and converted any oxide to fluoride. The graphite components were used in contact with the fluoride melt to prevent corrosion and dissolution of the metal pot into the fluoride salt melt. The hydrogen fluoride flow rate was determined by titration of the exit gas with standardized NaOH solution. The first salt mixture was hydrofluorinated for 8 h with an average flow rate of 32 mL/min hydrogen fluoride. The second salt mixture was hydrofluorinated for ~20 h with an average hydrogen fluoride rate of 30 mL/min. Since the appearance of the first sample indicated that some oxide may still be present after hydrofluorination, the time of hydrofluorination was increased for the second sample. (Previous experience has established that this amount of sparging is ordinarily effective in reducing the oxide content of the salt to <1 ppm).

After hydrofluorination, both salt mixtures were sparged with helium (while still molten) to free the salt of dissolved HF and were then allowed to solidify. The entire vessel assembly was put into the controlled atmosphere glove box where the solidified salt plugs were removed, scraped of surface graphite film that migrates to the salt interface and stored for irradiation. It is important to remove all graphite from the salt in order to be certain that no graphite/fluorine reactions occur unless intended.

2.2.2 *Fluoride salt irradiations* The pulverized fluoride salts were irradiated at the High Flux Isotope Reactor (HFIR) at ORNL. The spent fuel elements which are stored in the reactor pool contain a sufficient gamma flux for irradiation of the salt samples in a reasonable time period. The gamma flux of the elements varies in the range of 1.0×10^8 to $< 1.0 \times 10^7$ R/h (depending on the age of the element) and has a reasonably uniform intensity profile within the middle 50 cm of the fuel element length.[10] The storage racks for the elements contain a 10 cm cadmium sleeve through the center hole of each element to act as a neutron absorber. One concern with using the spent fuel elements was that the heat generated by absorption of radiation from the gamma source might produce too high a temperature in the sample. Prior to the irradiation experiments, a lead weight with two thermocouples embedded was used to measure the temperature of a spent fuel element over a 15 day period beginning with a 3-day-old spent fuel element. The temperature due to gamma heating dropped from 58 to 41°C over this period but appeared to level somewhat after 15 days in the range 41 to 38°C. The spent fuel elements were generally used within this 45-day period after being removed from the reactor, and the temperatures in this range were judged to be acceptable.

A nickel vessel with a conflat flange top and soft-copper gasket seal was fabricated for use as an irradiation vessel (Figure 1). The vessel, associated tubing, and gauge were passivated with fluorine prior to loading with a fluoride salt mixture to prevent reaction of the fluorine liberated during the actual irradiation experiments with these components. The loaded vessel was lowered into the center of a HFIR spent fuel element for irradiation. A 3.2 mm nickel tube and thermocouple wire extended to the top of the pool (~ 6 m) to an appropriate pressure gauge and temperature indicator. The reactor pool water which circulated around the vessel provided sufficient cooling during the course of the irradiation. Pressure and temperature readings were recorded on a daily basis over the period of the irradiation.

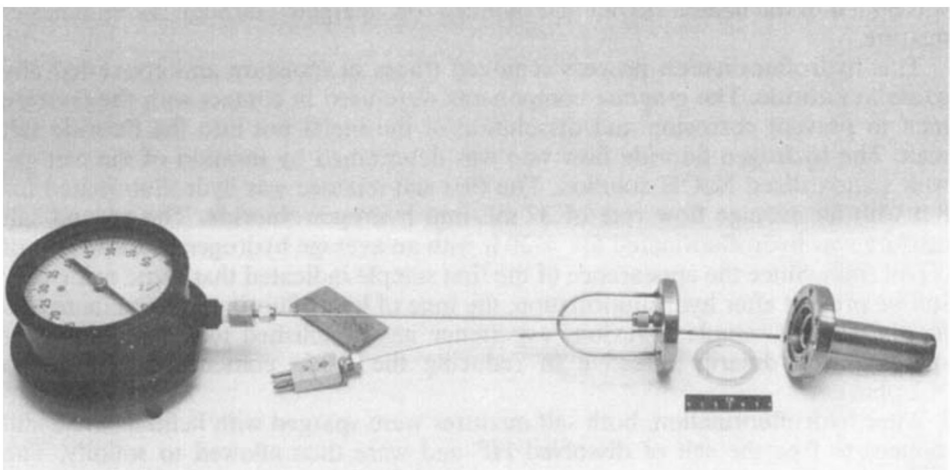

FIGURE 1 HFIR irradiation vessel with pressure gauge for irradiation of MSR salts. The connecting 1/8 inch nickel tube was replaced with one which was actually 20 ft long.

No direct analysis of the gas space was performed due to the limited nature of this work. However, previous work[8] has already established the identity of the gas evolved from the fluoride salt and our observations, upon venting the overpressure (cloudy, hydrolyzing gas emanating with the sharp, pungent odor of F_2), were consistent with what the previous researchers have reported.

The volume of the empty vessel was 40.0 cm^3 and was loaded with 30 g salt (12.3 cm^3, since the salt density is 2.43 g/cm^3) which yielded a net gas volume of 27.7 cm^3 in the vessel. This gas volume, coupled to that of the connecting tube and gauge (47.7 cm^3), gave a total gas volume of 75.4 cm^3. The gas/salt volume ratio was therefore: $75.4/12.3 = 6.1$.

In order to examine the effect of temperature on the salt irradiation, a "dry tube" was used which consisted of an ~ 10 cm diam stainless steel tube which extended from the top of the reactor pool to the center of a spent fuel element. The temperature in this tube was typically at 70°C (due to the absence of the cooling influence of the HFIR pool water). With this arrangement the nickel vessel containing the salt was lowered into the tube and center of a spent fuel element.

A second (and more satisfactory approach) was later developed in order to permit controlled heating of the nickel vessel. A nickel can was fabricated to enclose the sample vessel in a water-tight environment and thus permit the latter to be resistively heated in a dry atmosphere. This was accomplished by sealing the sample vessel inside the nickel can by soft soldering the lid and tubing lead-throughs (Figure 2). The electrical connections for the heating tape extended through a 6.4-mm tube to the top of the pool where a variac was used as a controller.

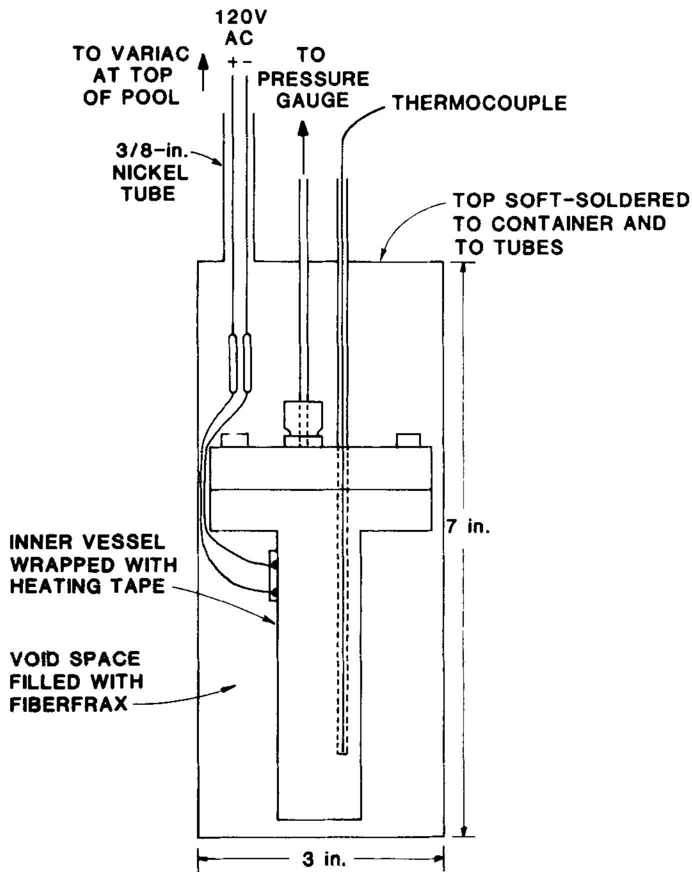

FIGURE 2 Schematic of nickel irradiation vessel in sealed, insulating container which was used for maintaining vessel at elevated temperatures.

3 RESULTS AND DISCUSSION

3.1 *Fluorine Generation*

As stated previously, in order to reach a steady-state condition with respect to radiolytic fluorine generation (i.e., one in which the generation rate was equivalent to the recombination with active sites in the salt), it would be necessary to have a very intense source of radiation. The previous study[8] using ^{60}Co gamma radiation at a dose rate of 0.45×10^{20} eV/h/g in a 35 g salt sample was achieved with a gamma flux rate of 0.72 MR/h. However, the experiments never reached a limiting fluorine pressure. Therefore, the HFIR[9] spent fuel elements, with a gamma flux of 10 to 100 times the above intensity, were considered essential in order to achieve the desired limits. Nevertheless, during the early stages of this work, an experiment on the local ^{60}Co irradiator[11] was performed—mainly to insure that calculated rates were correct and controlled experiments were possible in both the irradiator and, especially, in the HFIR spent fuel element. The experiment in the ^{60}Co irradiator showed little, if any, fluorine pressure increase over a one week period when exposed to a gamma flux rate of ~0.6 MR/h. Therefore, the much higher fluxes expected in the HFIR fuel element would not produce disastrous effects.

Several irradiation experiments in HFIR fuel elements were performed at approximately 40°C. The first of these involved 30 g of MSR composition salt in the nickel irradiation vessel. An induction period of about 6 days was observed where there was a slight decrease in the total pressure in the vessel followed by a slow increase in the pressure [0.65 kPa/d (0.09 psig/d) to a limit of 12.4 kPa (1.8 psig)] that continued for 18 days following the induction period. Because a higher rate of fluorine generation was expected (by extrapolating that previously reported[8] for the fluxes in the ^{60}Co irradiator to values as found in the HFIR spent fuel elements), the experiment was terminated to examine the contents for any problems. None were found.

A second experiment was begun to repeat the first one with more thorough fluorine passivation of the container, since it was suspected that the slight drop in pressure during the induction period might be due to further passivation of the container in the radiation field. The results of this experiment are shown in two figures (Figures 3 and 4) which present the data in terms of fluorine pressure as a function of days in the spent fuel element. The first of these figures, Figure 3, shows the fluorine pressure as it was actually measured with a 205 kPa (15 psig) pressure gauge. When the fluorine pressure reached the 205 kPa limit, it was bled off to ~136 kPa (5 psig) and allowed to accumulate again. In this manner, the sawtooth appearance of the pressure/time plot was produced. The pressure continued to build up at a steady rate of 0.19 kPa/h (0.028 psi/h) until after ~100 day, it leveled off.

The same data are presented in Figure 4, but this time the net pressure as a function of time was plotted. In this figure, it is more obvious how the fluorine generation progressed without the necessary bleed-off operation. A rather constant rate of generation up to 446 kPa (50 psig) limiting pressure was observed. It is apparent that the bleed-off procedure had little effect on the rate of generation or the limiting value of 446 kPa, since the rate did not measurably change as a result of bleed-off. Furthermore, it has been suggested[5] that these factors are more controlled by the concentration of defects within the salt lattice itself than by conditions external to the crystal lattice. Therefore, it is not expected that the changes in the F_2 pressure over the crystal should have any measurable effect on the rate of fluorine generation. Instead, the F_2 generation and recombination rate should be controlled by the concentration of defect sites in the crystal lattice with which the halogen can combine.

The gamma flux for the particular fuel element is given in Figure 4 to demonstrate the effect of flux on the rate of fluorine generation. As shown by the discontinuities in this flux curve, several fuel elements were used for the irradiation in an effort to maintain the maximum possible flux. The gamma flux of the spent fuel elements was recently reported.[10] Even though there was a noticeable change

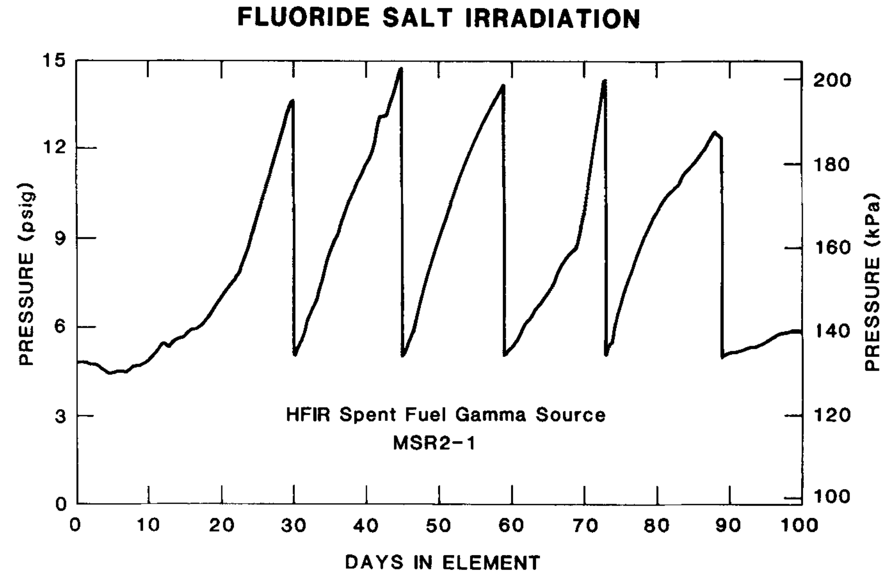

FIGURE 3 Fluorine pressure rise over irradiated MSR salt as a function of exposure time in the HFIR fuel element.

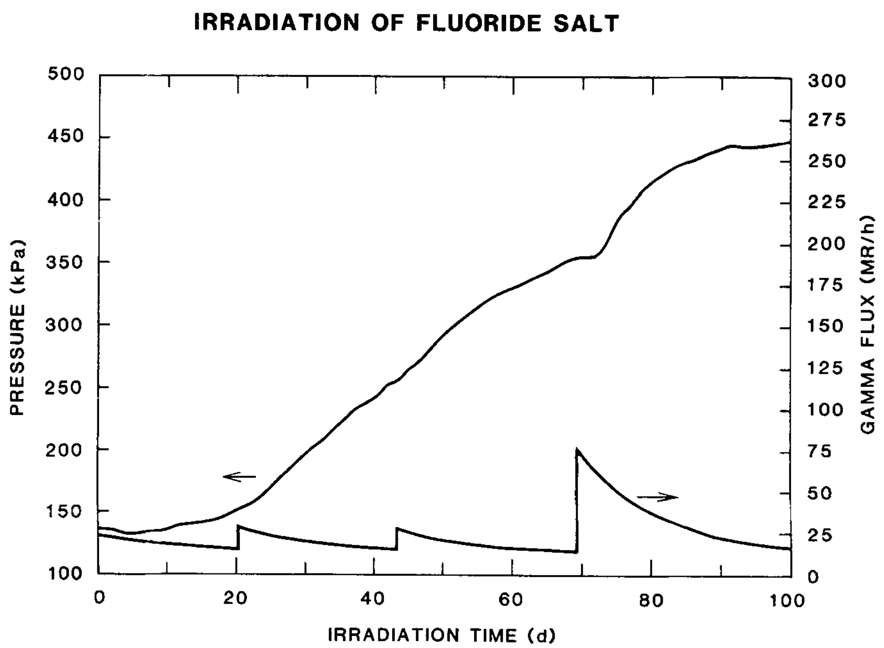

FIGURE 4 Net fluorine pressure accumulated over the irradiated MSR fuel salt and gamma flux rate as a function of time in the fuel elements. The latter shows the effect of four different fuel elements used during the irradiation period.

in the generation rate when the more intense source was used (cf., 70 days' time in Figure 3), it is not a linear relationship, thus confirming the earlier observations that the rate was not 10 to 100 times greater in the HFIR fuel element than in the ^{60}Co source even though the fluxes varied by that amount. Furthermore, the rate of pressure buildup at 100 days had decreased relative to the rates just prior to the other fuel element changes (cf., growth rates at 20 and 43 days with that at 100 days), thus demonstrating that a limiting pressure had been reached.

The damage to the crystal lattice was computed by dividing the total moles of fluorine appearing in the sample gas space, 75 cm^3, by the total moles of fluorine available from the 30 g of salt. The limiting value of approximately 2% damage is consistent with that which has been reported[1] for other halide salts using high doses of neutron irradiation; but it is 100 times greater than the point defect saturation level using gamma rays of lower intensity or 400 keV electrons.[2] This steady-state limit represents the point where the number of defect centers (containing active metals such as Li or Be) are so great that the rate of molecular or atomic fluorine recombination with them is equal to the rate of fluorine atom generation from gamma radiolysis of the fluoride salt. Therefore, in the higher radiation field, such as the HFIR spent fuel elements where the rate of F_2 generation is much faster, the total amount of radiation damage at steady state should be correspondingly greater.

The 446 kPa pressure limit is also dependent on the volume of the gas space above the irradiated salt because of simple equilibrium expressions. If the volume over the salt had been larger, then the limiting pressure would have been smaller. Conversely, fluorine pressures as high as 50 atmospheres (5.05 MPa) have been reported[8] when the gas/salt volume ratio was extremely small. In addition to these volume ratio effects, the determining factor on the limiting fluorine pressure appears to be most dependent on the concentration of defect sites produced. When the defect sites reach approximately 2% of the salt content for the radiation flux used here, no more generation and release of fluorine should occur.

Attempts to reproduce these data were only partially successful. It was impossible to leave the irradiation vessel undisturbed during the long 100 day term of an experiment because necessary changes in the spent fuel elements must be made. As a consequence of the handling, leaks frequently appeared in the system and compromised the validity of the results. The final attempt to measure the limiting fluorine pressure was interrupted (because of a reactor shutdown and resulting disruption in the supply of spent fuel elements) after the fluorine pressure had reached 8 psig (156 kPa).

Similar irradiations of fluoride salt samples at higher temperatures—namely in the 70 to 100°C range were conducted. No accumulation of F_2 over the salt at these temperatures was ever observed. Even though the F_2 recombination rate is expected to equal that of the generation at 150°C,[8] these results indicate that the recombination rate had approached that of generation at the lower temperatures. Nevertheless, these experiments at higher temperatures should be repeated before a definite statement regarding temperature effect can be made.

3.2 *Recombination Studies*

After the fluorine pressure reached the limiting value in the above experiment, the recombination characteristics of the fluorine with the salt were studied. The sample pot and associated tubing and gauge was removed from the fuel element radiation source and heated for a period of time at increasing temperatures while monitoring the fluorine pressure. These results are shown in Figure 5 for data taken at 100, 150 and 200°C. An activation energy of 39 kJ/mol for recombination was computed from the slope of the line, which is slightly less than half the value calculated from Haubenreich's report.[12] However, the significant difference between our experiment and those discussed by Haubenreich was in the form of the salt from which fluorine evolution and recombination were measured. Our experiment used a pulverized salt sample (to minimize fluorine diffusion within the salt) whereas Haubenreich reports on the fluorine evolution from a single piece of solidified salt (of comparable size to our sample). Consequently, the difference in the activation energies most probably reflects the diffusion characteristics of fluorine within the salt lattice; and then, in a practical sense, these two extremes in the activation energy reflect the range that would be expected for fluorine recombination, depending upon the physical conditions of the salt. Even though a finite rate of recombination at 40°C should exist, a value of only 3.8 Pa/h was calculated from the least squares fit of the data. These data further predict that the temperature at which generation would equal recombination is 150°C-a value which was suggested in the earlier MSR experiments.[8]

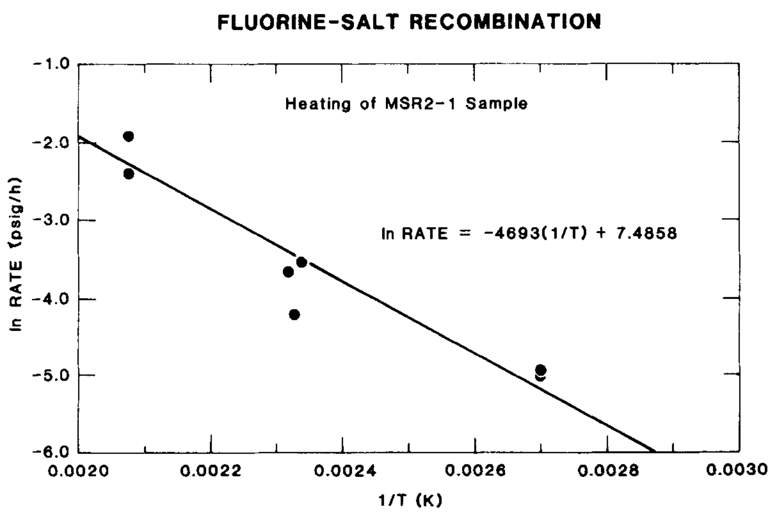

FIGURE 5 Rate of fluorine recombination with fuel salt as a function of temperature during the recombination tests.

4 CONCLUSIONS

It has been shown that when fluoride salts are exposed to large doses of gamma radiation fluorine gas is liberated from the crystalline lattice and appears as a gas overpressure. The maximum limit of damage to the crystalline material is 2 mol%, as determined by the amount of fluorine gas released. At that point, the rate of F_2 recombination with active metal centers is equal to the rate of F_2 generation. The F_2 overpressure will react with the active sites in the crystalline salt if the temperature of the system is elevated to temperatures of 100 to 300°C. But measuring the activation energy of the recombination reaction, it is evident that the rate of recombination should equal the rate of generation at 150°C under these conditions of irradiation. If the dose rate were lower, as in earlier work, this temperature would be expected to be much lower.

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